Generative Adversarial Networks

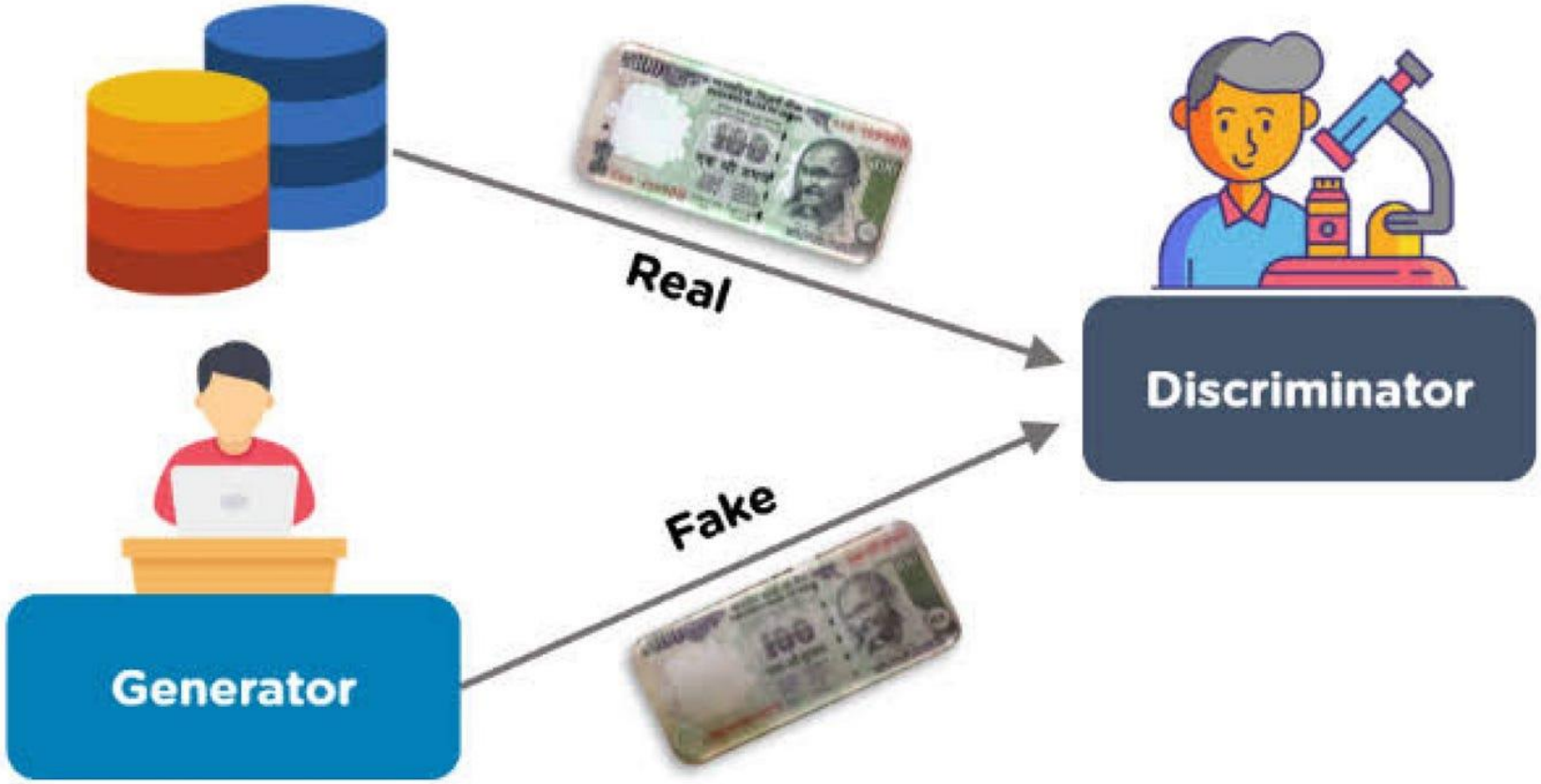
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Introduction

Deep learning models designed to generate new data by learning patterns from datasets

Introduced in 2014 by Goodfellow et al.

GANs consist of two neural networks which compete adversarially

- Generator
- Discriminator



https://arxiv.org/pdf/1406.2661

Not require labeled data (unsupervised learning)

Can generate highly diverse and realistic output (images, video, text, audio...)

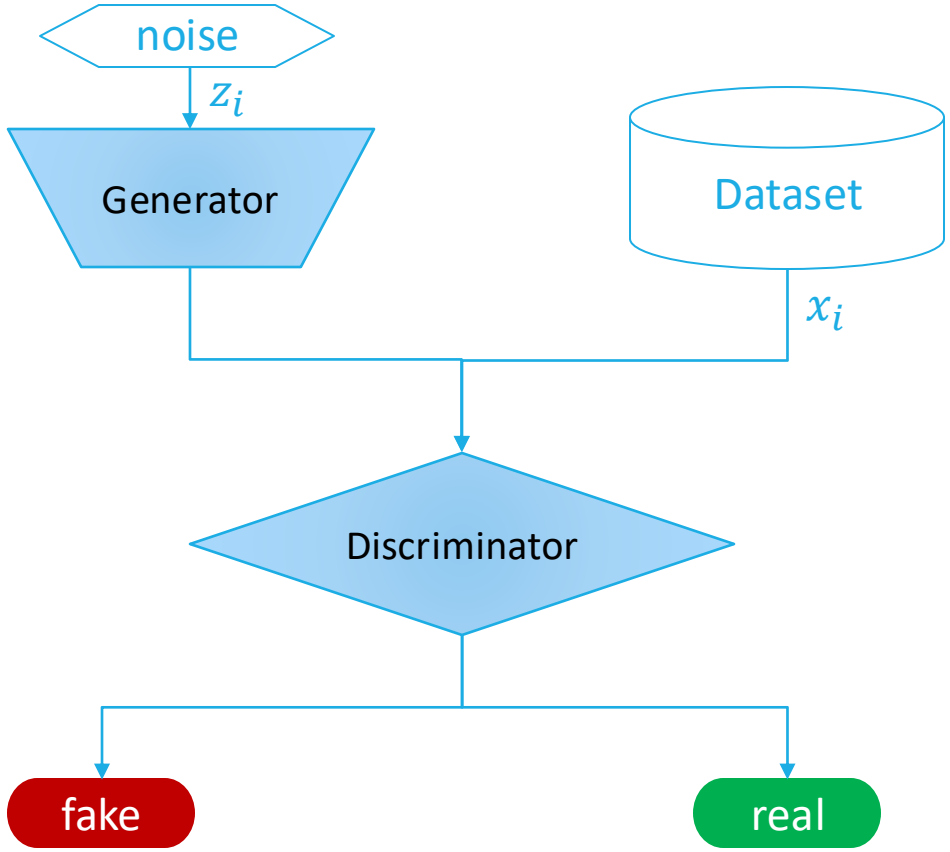
Require extensive computational resources and careful tuning to avoid training instabilities

Architecture

Main components

- Generator
  - Produces new samples based on input noise and learned patterns
  - Goal: Maximize the cross-entropy when discriminator fail
- Discriminator
  - Classifies samples as real or fake
  - Goal: Minimize the binary cross-entropy
  - $Loss = -\frac{1}{n}\sum_i^n y_i \log(D(v_i)) + (1 - y_i)\log(1 - D(v_i))$

$$-\frac{1}{n}\sum_{i}^{n} \log(D(x_i)) + \log\big(1 - D(G(z_i))\big)$$



Training

Involves a minimax game where the Generator tries to fool the Discriminator, and the Discriminator tries to correctly identify generated data

- $\min_D \max_G \left\{ -E_x[\log(D(x))] - E_z\left[\log\left(1 - D\left(G(z)\right)\right)\right]\right\}$

Continuous improvement in data generation quality

Until the Generator produces highly realistic outputs

The goal is to reach equilibrium, where the Discriminator cannot distinguish fake from real data



Evaluation metrics

Inception Score (IS)

- Measures quality and diversity of generated images
- Uses a pretrained classifier (Inception)

Fréchet Inception Distance (FID)

- Compares statistics of generated images to real images
- Lower values indicates closer match

Mode Score and Mode Collapse detection

- Quantifies diversity and detects missing modes in generated data



In Proceedings of the 2024 7th International Conference on Computational Intelligence and Intelligent Systems (CIIS '24). Association for Computing Machinery, New York, NY, USA, 79–85.

Variants

Vanilla GAN

- Original and basic framework for all GAN variants

Conditional GANs (cGANs)

- Controlled data generation by adding conditioning variables (labels)

Deep Convolutional GANs (DCGANs)

- Image generation (realistic)

CycleGANs

- Image to image translation without paired data

StyleGANs

- High quality, controllable image synthesis



Practice 5

Creating a simple GAN

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